AnchorMTM: Spatially Anchored Merge Tree Maps with Position Bounds

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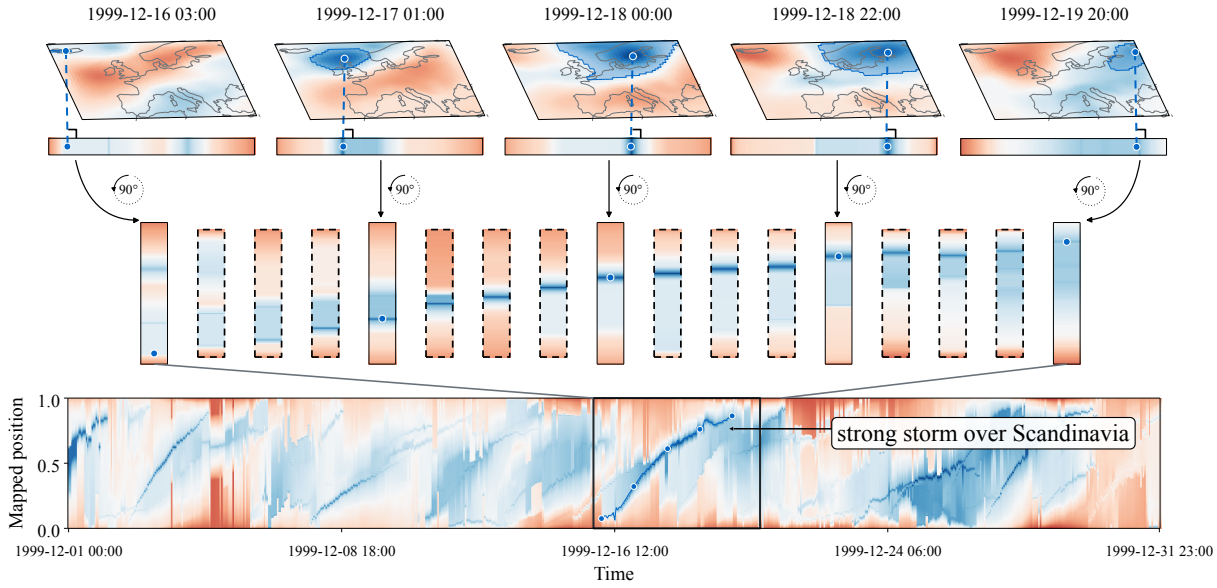


Figure 1: Qualitative Storms archive ($\varepsilon = 5\%$; preprocessing differs from Table 3). Spatial snapshots and profiles identify one matched extremum; blue outlines show leaf-arc supports. Rotated profiles and dashed intermediate samples connect to its trajectory in the full 744-frame map.

Abstract

Temporal merge tree maps show how the topology of a scalar field evolves in one image, but their display axis has no fixed spatial meaning, so a feature's position cannot be read as its location. Anchoring the axis to one spatial direction collides with the hierarchy: an exact profile keeps every subtree contiguous, while the projections of sibling subtrees may interleave. How much displacement this conflict forces, and where, has not been quantified. We prove three lower bounds on the displacement that exact topology, finite feature widths and a bounded merge error force on every such layout. The bounds depend only on the tree, the projected feature locations and the widths, so one computation applies to any map: it locates the merges that force the displacement and the merge error that releases it, and it separates the error of a given map into a part forced by the hierarchy and a part chosen by its layout. AnchorMTM, a spatially anchored construction with one control, a merge-error budget ε, approaches the bounds by a leaf-order search and a capped quadratic program. The bounds equal the exhaustive optimum on controlled trees; on 606 of 614 frames of five planar sequences, the selected exact-topology order attains the width-aware bound, while 31–95% of the order error of TMTM and ST-MTM is not forced by the tree. On four planar sequences of the ST-MTM benchmark, the mean worst displacement of AnchorMTM is 73–96% lower in Exact Mode and 77–98% lower with a selected budget than for same-input TMTM and ST-MTM implementations, every rendered map stays within its budget, and ST-MTM's published neighborhood preservation remains higher.

Index Terms: Time-varying scalar fields, merge tree maps, topology preservation, spatial fidelity, lower bounds, constrained layout.

1 Introduction

An analyst who follows storms across an ocean or fire activity across a region reads a temporal overview for three things at once: which features exist and how they nest, where they are, and how they move. Animation keeps the spatial view of each instant but makes distant times hard to compare; static overviews give one display axis to time and compress space into the other [1, 29, 32]. What that second axis means decides which of these questions the overview can answer. Figure 1 links the spatial supports of one feature to such a map.

Merge tree maps answer the first question well. Temporal Merge Tree Maps (TMTM) draw the merge tree of each frame as a 1D scalar profile and align the profiles along time, so minima show extrema and the barriers between them show merge levels [21]. Spatiotemporal Merge Tree Maps (ST-MTM) add relative spatial distances and temporal position constraints to the layout [9]. Neither gives its display axis a fixed spatial meaning: TMTM orders leaves for the tree and for temporal alignment, and pairwise distances fix relative positions but no origin or direction. A feature's position therefore cannot be read as its longitude or its distance along a flow, and a feature that moves east need not move on the map. Hovmöller diagrams take the opposite side: they keep one physical coordinate and average the field over the others, so nearby extrema blend and

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the hierarchy disappears [20].

Anchoring the axis means projecting every extremum onto one direction and placing it near its projection. The merge tree resists. An exact profile keeps every subtree in one contiguous block, but the projections of sibling subtrees can interleave. In Figure 2, four minima project in the order A, C, D, B, which separates A and B by the C, D subtree, so some features must leave their references or some merge levels must change; finite feature widths add packing constraints. ST-MTM observes that a one-dimensional layout cannot preserve all spatial distances, so some distortion is an expected consequence of the projection [9]. How much of it the hierarchy forces, where it arises and how much merge error buys it back have not been quantified, yet they decide whether a spatial axis is worth having.

We answer these questions. Three lower bounds give the displacement that exact topology, finite feature widths and a bounded merge error force on every admissible layout. They depend on the tree, the references and the widths, not on a solver, so the same computation measures any map, including TMTM and ST-MTM, and separates its error into a part forced by the hierarchy and a part chosen by its layout. AnchorMTM, *Spatially Anchored Merge Tree Maps*, then builds maps that approach the bounds. It estimates one reference direction from matched extremum motion or takes one from the analyst, searches for a leaf order within a merge-error budget ε, sets optimal merge barriers, places anchors with a capped quadratic program and reconstructs the scalar profiles on a shared temporal axis. Exact Mode ($\varepsilon = 0$) keeps the merge tree; Budgeted Mode ($\varepsilon > 0$) trades a declared merge-level error for position.

Our main contributions are as follows:

- a formulation of spatially anchored merge tree maps as hierarchy-constrained interval layouts with a fixed reference and a merge-level allowance, with three layout-independent lower bounds on the displacement that exact topology, finite widths and a bounded merge error force on any such layout, which locate each conflict and the merge error that releases it (Section 4);
- AnchorMTM, a spatially anchored construction with one primary control ε, built from a motion-derived reference, topology-budgeted leaf ordering, optimal merge barriers and capped spatiotemporal anchor placement (Sections 3.3 and 5);
- an evaluation showing that the bounds are attained on controlled trees and by the selected order on 606 of 614 planar frames, that 31–95% of the order error of TMTM and ST-MTM is not forced by the tree, and that AnchorMTM places features much closer to their references than TMTM and ST-MTM on the four planar sequences of the ST-MTM benchmark and two volumetric flows, within every declared merge budget (Section 6).

2 Related Work

2.1 Spatially Referenced Temporal Visualization

Static temporal displays expose distant observations simultaneously, while animation retains the spatial view of each instant; their usefulness depends on the analytical task [29, 32]. Generalized space–time cubes describe the transformations connecting such views [1]. In compressed temporal overviews, the spatial coordinate determines which location and motion relationships remain directly interpretable.

Hovmöller diagrams retain a selected physical coordinate and summarize the other dimensions [20]. Geographic orderings and space-filling curves instead traverse the domain [8, 16]. Dynamic Volume Lines and data-driven curves extend spatial traversal to volumetric comparison and scalar-aware ordering [36, 38]. A traversal can retain spatial neighborhoods, but its 1D coordinate is generally different from a fixed projection, and traversal adjacency alone does not enforce the merge hierarchy.

1D projections organize spatiotemporal phenomena [14], and MotionRugs expose collective behavior of moving entities [5]. More generally, an embedding trades original-domain relationships against its display organization [26]. Relative proximity does not determine an absolute coordinate. AnchorMTM assigns fixed projection targets to topological features, and we bound their displacement under hierarchical and interval constraints for any layout, combining spatial reference with a temporal topology overview.

2.2 Topology-Based Temporal Maps and Geometry-Aware Layouts

Topology-based visualization organizes scalar structures through connectivity [17]. Landscapes and landscape profiles express hierarchies through terrain or 1D scalar representations [27, 34], with geometry-preserving and temporally coherent extensions [3, 18]. These representations connect scalar values and feature relationships in a compact view.

TMTM emphasizes intact topological features and temporal alignment in static merge-tree maps [21]. ST-MTM incorporates relative spatial geometry and temporal position constraints [9]. These priorities give spatial relationships and coherence a role in the map, while leaving its display coordinate free to differ from a fixed physical direction. AnchorMTM estimates a shared direction from extremum motion, or accepts a physical direction, and couples that reference to topology-budgeted layout and to position bounds that hold for any layout. It reuses the established profile and scalar-filling foundation.

Hierarchy-constrained leaf ordering traditionally minimizes adjacent dissimilarity [2]; target-order reordering studies related tree-consistent permutations [6]. Separation-constrained layouts control geometric overlap and displacement [11, 23]. AnchorMTM couples these layout concerns to fixed spatial targets, and our bounds state the displacement they force. Its exact-topology point-order bound has a closed form, whereas finite-width candidate ordering remains heuristic.

Merge-tree comparison depends on the descriptors and correspondences used [37]. Labelled cophenetic/interleaving comparisons support fixed-identity merge-level accounting [15, 25], and ordered merge trees connect scalar curves to monotone interleaving distances [4]. We use fixed labels to measure displayed merge error. The optimal-barrier subproblem relates to minimax ultrametric fitting [13], with the additional restriction that one linear order induces every displayed merge.

3 Spatially Anchored Merge Tree Maps

We first describe merge tree maps in one formulation that covers the existing ones, then give the axis a spatial reference and show where it conflicts with the tree.

3.1 Merge Trees and One-Dimensional Profiles

For a scalar field f_t on a 2D or 3D domain, the merge tree T_t records how sublevel-set components appear at minima and join at saddles as the threshold increases; negating the field gives the split tree [17, 31]. Persistence simplification precedes layout, and all statements refer to the simplified tree [12]. Each leaf i is an extremum at location p_i^t; its support S_i^t is the set of samples on its leaf arc, with measure m_i^t. Two leaves merge at the level $V_{ij}^t = h_t(\mathrm{lca}(i, j))$ of their lowest common ancestor, and V_{ii}^t is the extremum value; equal-level merges form one multifurcation.

A one-dimensional profile draws the leaves as minima in a chosen leaf order, each inside an interval sized by its support, and fills the gaps from the tree, so that the largest profile value between two minima shows their merge level [21, 27, 34]. The profile shows the tree exactly only if every subtree occupies one contiguous block of

leaves. Stacking the profiles along time gives a temporal merge tree map; leaves of consecutive frames are matched by support overlap with a one-to-one assignment, and $\mu_t(i)$ denotes the matched predecessor of leaf i [9].

3.2 A Common Formulation

A map frame consists of a leaf order π, intervals with centers z_i and widths W_i, display anchors x_i for the extrema, and a profile ψ. The layout set $\mathscr{L}(\pi;W)$ contains the placements whose intervals follow π on the canvas $[0,L]$, are disjoint with gap at least g, and hold each anchor within $\rho W_i/2$ of its center. Two errors describe a frame. The *merge error* compares displayed and true merge levels of fixed labels,

$$d_{\mathrm{top}}(\psi,T) = \max_{i,j} \left| V_{ij}^{\psi} - V_{ij} \right|, \tag{1}$$

where V_{ij}^{ψ} is the largest profile value between the anchors of i and j. It is zero exactly when the profile shows the tree, and for complete labels it equals the labelled interleaving distance, which bounds the interleaving and bottleneck distances [15,24,25]. The *positional error* compares anchors with spatial targets q_i (Section 3.3); its worst value $\|x-q\|_\infty$ is reported per frame. Two order-level versions isolate the role of the leaf order:

$$\begin{aligned} \mathrm{NROE}(\pi) &= \frac{1}{2L} \max_{j \le k} \left[q_{\pi(j)} - q_{\pi(k)} \right]^+, \\ \tau(\pi) &= \min_{(x,z)\in\mathscr{L}(\pi;W)} \|x-q\|_\infty. \end{aligned} \tag{2}$$

NROE is the smallest worst displacement of points placed in order π, relative to L; τ adds the intervals and is computed exactly for a fixed order.

Each map we consider solves, frame by frame, an instance of

$$\begin{aligned} \min_{\pi\in\Pi,\ (x,z)\in\mathscr{L}(\pi;W)} \ & \lambda_R E_R + \lambda_G E_G + \lambda_M E_M + \lambda_C E_C \\ \text{s.t. } & d_{\mathrm{top}}(\psi,T) \le \eta. \end{aligned} \tag{3}$$

Here E_R is the mean squared distance of anchors to their references, E_G a stress between anchor separations and support-centroid distances, E_M the mean squared distance to a temporal target $\hat{x}_i$ derived from the matched previous anchor, and E_C the offset of anchors from interval centers. A method is a choice of order rule Π, weights λ, temporal target $\hat{x}$, width semantics W and allowance η (Table 1). TMTM and ST-MTM keep the tree ($\eta = 0$) and have no reference term ($\lambda_R = 0$): TMTM orders children for temporal overlap and packs intervals by sample count [21], and ST-MTM orders leaves within the tree for small adjacent distances and fits relative distances with a stability target [9]. Their axes therefore encode tree traversal or relative distance rather than a location. Projections and space-filling curves sit at the other end: they follow space and ignore the tree [14]. AnchorMTM adds the reference term and a declared allowance η, controlled by one number ε (Section 5). The correspondence concerns the problem each method solves; solvers differ in scaling and other details, and Section 6 evaluates implementations of the published methods, not our solver with their weights.

3.3 Spatial Reference

A reference gives the axis a meaning: $q_i^t = \langle a, p_i^t \rangle - o$ for one unit direction a and one origin o, where o and the extent L come from the projected corners of the domain. By default, a is the direction of largest matched extremum motion,

$$a^* \in \arg\max_{\|a\|=1} \ a^\top \Big(\sum_{t,i} \Delta p_i^t (\Delta p_i^t)^\top \Big) a, \tag{4}$$

Table 1: Maps of time-varying scalar fields as instances of Eq. 3. Order rule: how leaves are ordered; placement: nonzero terms and temporal target; η: allowed merge error relative to the scalar range (tolerance ε_0 omitted); axis: what a position encodes.

Method	Order rule	Placement	η	Axis
TMTM [21]	children by overlap with the previous frame	packed by sample count; no optimization	0	tree traversal
ST-MTM [9]	tree-constrained OLO, kept unless cheaper by a ratio	λ_G (stress), λ_M with $\hat{x}$ = previous anchor	0	relative distance
OLO [2]	tree-constrained OLO	ranks only	0	order
Projection, Hilbert, Morton [14]	by reference or curve	at reference or curve rank	none	location or curve rank
AnchorMTM	least τ within the budget	λ_R dominant, $\hat{x}$ follows projected motion	ε	location

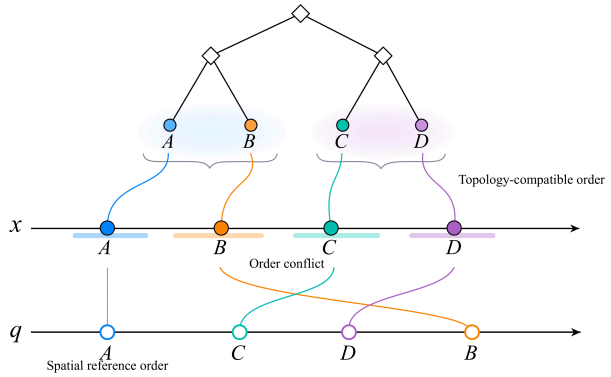


Figure 2: Reference–topology conflict. The tree requires contiguous A,B and C,D blocks, but the references occur as A,C,D,B. Any exact profile must displace some anchors (colored links); the bounds of Section 4 give the least such displacement. Widths and positions are schematic.

with $\Delta p_i^t = p_i^t - p_{\mu_t(i)}^{t-1}$ over valid matches, signed so that the total projected motion is nonnegative; a direction chosen by the analyst can replace it. Direction, origin and extent are fixed for the whole sequence and all budgets, with no per-frame recentering, so a feature that moves along a moves along the axis. Matching by support overlap can miss fast features, which then contribute no motion sample.

3.4 Where Topology and Space Conflict

An exact profile keeps every subtree contiguous, but the references of sibling subtrees can interleave. In Fig. 2 the references occur in the order A,C,D,B, which separates A and B by the C,D subtree, so no exact profile places all four anchors at their references; finite widths add packing constraints. How large must the displacement be, where does it arise, and how much merge error removes it? The answer depends only on T, q, W and η, so it holds for every instance of Eq. 3 at once.

4 The Positional Price of Keeping the Tree

The three bounds below involve the tree, the references, the widths and the allowance, but neither the order rule nor the weights, so they hold for every row of Table 1. The first concerns exact topology, the second adds widths, and the third allows a merge error. Full proofs

are in Supplementary Material; we drop t within a frame, write $[v]^+ = \max(v,0)$ and take empty maxima as zero.

4.1 Exact Topology

On the reduced tree, where equal-level merges form one node, exact topology makes every subtree contiguous. Let $[m_X, M_X]$ be the reference range of the leaves of subtree X.

Theorem 1 (Exact-Topology Position Bound)**.** *Every profile with $d_{\text{top}} = 0$ has $\text{NROE}(\pi) \geq \nu^*(T,q)$, where*

$$\nu^*(T,q) = \frac{1}{2L} \max_{v} \max_{\substack{X,Y \in \text{ch}(v) \\ X \neq Y}} \big[\min(M_X - m_Y, M_Y - m_X)\big]^+. \quad (5)$$

Ordering children recursively by the midpoints of their reference ranges attains ν^* in $O(n \log n)$ time.

Proof sketch. Two sibling blocks must take one of two orientations, with inversion costs $[M_X - m_Y]^+/2$ and $[M_Y - m_X]^+/2$; midpoint order takes the smaller, and every inverted pair of leaves is separated by sibling blocks at its lowest common ancestor. □

The maximizing node and sibling pair say where the price arises, at the merge of two subtrees whose reference ranges overlap, and the value says how large it is: half the overlap left in the cheaper orientation.

Forced and chosen error. Because midpoint ordering attains ν^* on the same tree, the order error of any exact-topology map splits into $\nu^*(T,q)$, which the tree forces, and $\text{NROE}(\pi) - \nu^*(T,q)$, which its order rule chooses and which can be removed without changing the tree. NROE does not involve widths, so the split applies equally to TMTM and ST-MTM, whose interval sizes follow different rules.

4.2 Finite Widths

Intervals need room: a block of leaves occupies at least the sum of its widths and gaps. For disjoint leaf sets X and Y, let $\omega(X,Y)$ be the least worst displacement that placing all of X before all of Y, or the reverse, forces on their anchors, whichever is smaller, and let E_∂ be the cost of fitting all intervals between the canvas ends. Both follow from sorted reference endpoints and cumulative widths (Supplementary Material).

Theorem 2 (Width-Aware Position Bound)**.** *Let $\mathscr{P}_0$ contain every pair of descendant-leaf sets of two siblings. Every exact-topology interval layout satisfies*

$$\begin{aligned} \|x - q\|_\infty \geq \tau(\pi) &\geq \mathscr{B}_w(T,q,W) \\ &= \max\Big\{E_\partial, \max_{(X,Y)\in\mathscr{P}_0} \omega(X,Y)\Big\}. \end{aligned} \quad (6)$$

With zero widths and gaps, $\mathscr{B}_w = L\,\nu^*(T,q)$.

Proof sketch. The first and last anchors of a block must enclose its intervals and gaps up to the anchor slack, which gives a shortfall against the reference separation; exact topology fixes one orientation of every sibling pair, and the canvas ends give the one-sided costs. □

$\mathscr{B}_w$ bounds every exact-topology order and need not be attained; $\tau(\pi)$ is the attained minimum for a given order.

4.3 A Bounded Merge Error

A merge allowance can release a block but still limits interleaving. If leaf k lies between two descendants i, j of node u, its displayed merge with i cannot exceed theirs, so $V_{ik} \leq h(u) + 2\eta$. Leaf k can therefore enter the block of u only if it descends from $A_\eta(u)$, the highest ancestor of u whose level is within 2η of $h(u)$. Let $\mathscr{P}_\eta$ contain the leaf-set pairs this rule keeps apart: $(\mathscr{D}(u), \{k\})$ for leaves k outside $\mathscr{D}(A_\eta(u))$, and $(\mathscr{D}(u), \mathscr{D}(w))$ for internal nodes u, w with $\mathscr{D}(w)$ disjoint from $\mathscr{D}(A_\eta(u))$ and $\mathscr{D}(u)$ disjoint from $\mathscr{D}(A_\eta(w))$, where $\mathscr{D}(\cdot)$ denotes descendant leaves.

Theorem 3 (Bounded-Topology Position Bound)**.** *In every profile with $d_{\text{top}} \leq \eta$, each pair in $\mathscr{P}_\eta$ lies wholly in one orientation. Hence*

$$\begin{aligned} \text{NROE}(\pi) \geq C_\eta &= \max_{(X,Y)\in\mathscr{P}_\eta} \frac{[\min(M_X - m_Y, M_Y - m_X)]^+}{2L}, \\ \tau(\pi) &\geq \max\Big\{E_\partial, \max_{(X,Y)\in\mathscr{P}_\eta} \omega(X,Y)\Big\}. \end{aligned} \quad (7)$$

The theorem also says what releases a conflict: a subtree may interleave with its sibling once 2η reaches the gap between its own merge level and its parent's, so conflicts at deep merges stay expensive. Flattening every parent gap of at most 2η gives a second valid bound; neither dominates, and we report the larger. Read in reverse, the theorem prices the other end of Table 1: a map whose order error is below C_η, such as a projection with zero order error, must show a merge error above η.

5 AnchorMTM: Approaching the Bounds

AnchorMTM is the instance of Eq. 3 with a reference term and the allowance $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$, where Δ_F is the scalar range of the sequence, $\varepsilon_0 = 10^{-7}$ a fixed numerical tolerance, and $\varepsilon \geq 0$ the only control. *Exact Mode* ($\varepsilon = 0$) keeps the tree up to ties within $\varepsilon_0\Delta_F$; *Budgeted Mode* ($\varepsilon > 0$) trades a declared merge error for position. Since Eq. 3 couples a discrete order with continuous placement, AnchorMTM solves it in two stages that follow the bounds (Fig. 3): it chooses the order by τ, the quantity the bounds constrain, and then places anchors within a cap on τ. Intervals have widths $W_i = s\,m_i$ with one fixed scale s; all other settings form one fixed configuration, listed in Supplementary Material.

5.1 Leaf Order within the Budget

For each frame, the order is chosen lexicographically,

$$\pi^\varepsilon \in \arg\min_{\pi\in\mathscr{C}} \big(\tau(\pi), \text{NROE}(\pi)\big) \quad \text{s.t.} \quad \delta^*(\pi) \leq \eta, \quad (8)$$

over a finite candidate pool $\mathscr{C}$: the exact tree ordered by midpoints (which attains ν^*), threshold and conflict-driven flattenings of the tree, and the orders visited while width-aware block moves refine the best of them. Every candidate is evaluated against the original merge matrix. The pool is heuristic, so the selected order is not a certified optimum over all permutations; Section 6 measures how often it attains the bounds. Admission uses the smallest merge error that an order can show.

Proposition 1 (Optimal Barriers)**.** *For a fixed order, let $C_k = \min_{i\leq k<j} V_{\pi(i)\pi(j)}$ be the lowest merge across gap k and F_k the larger of its two adjacent extremum values. The smallest attainable merge error is*

$$\begin{aligned} \delta^*(\pi) = \max\Big\{0,\ &\max_k (F_k - C_k), \\ &\tfrac{1}{2}\max_{i<j}\big(V_{\pi(i)\pi(j)} - \max_{i\leq k<j} C_k\big)\Big\}, \end{aligned} \quad (9)$$

attained by the barriers $b_k = C_k + \delta^*(\pi)$.

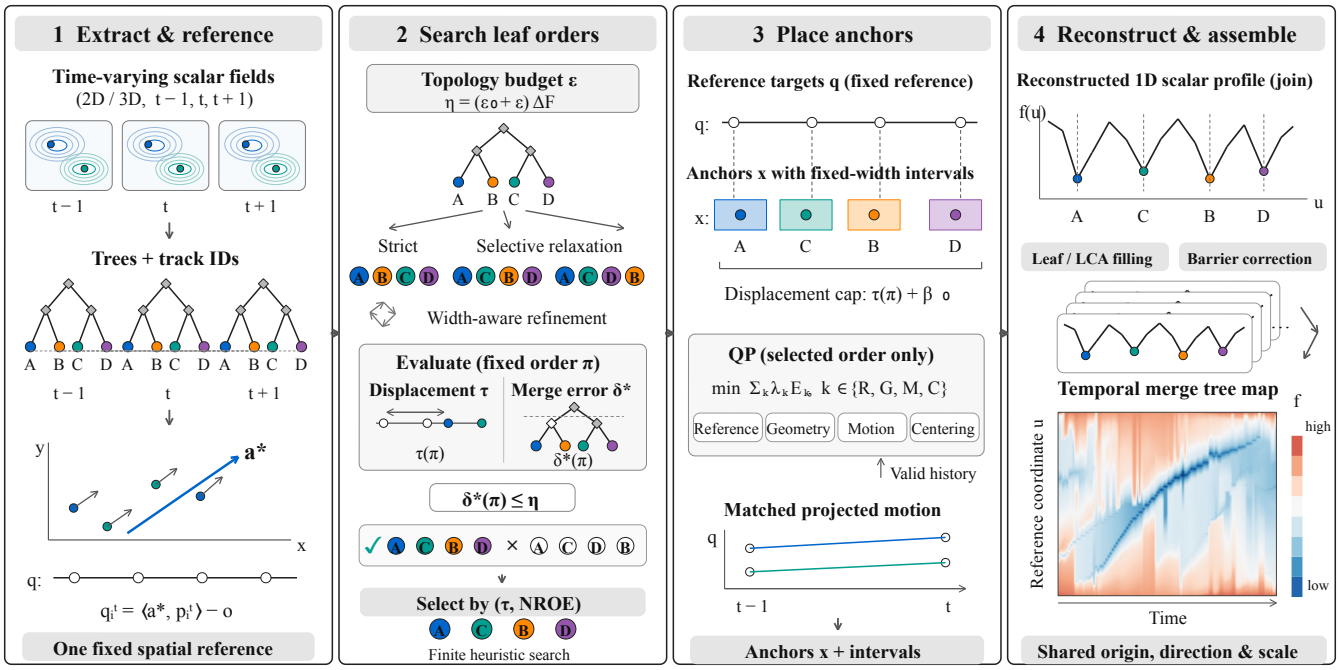


Figure 3: AnchorMTM construction. (1) Extract merge trees, match leaf supports and estimate one fixed reference from matched extremum motion. (2) Search leaf orders, admit those within the merge-error budget, and select by τ and NROE. (3) Place anchors within nonoverlapping intervals under a displacement cap, using matched history. (4) Reconstruct the profiles with corrected barriers and assemble the map. Diagrams are schematic; the map is a qualitative Storms output at $\varepsilon = 10\%$.

An order therefore meets the budget exactly when $\delta^*(\pi) \le \eta$, which makes admission a closed-form test.

5.2 Anchor Placement

For the selected order, AnchorMTM solves Eq. 3 with default weights $(\lambda_R, \lambda_G, \lambda_M, \lambda_C) = (4, 1, 0.5, 0.1)$, the temporal target $\hat{x}_i^t = x_{\mu_t(i)}^{t-1} + (q_i^t - q_{\mu_t(i)}^{t-1})$, which follows projected motion and keeps $x - q$ continuous, and the additional cap $\|x - q\|_\infty \le \tau(\pi) + \beta_0$ with a fixed slack β_0. The cap is feasible because τ is attained, and the problem is a convex quadratic program for a fixed order. The program cannot replace the selected order, so the cap transfers the order's positional guarantee to the anchors. Leaves without a matched predecessor have no temporal term.

5.3 Profile Reconstruction and Checks

Profiles reuse the leaf-arc sampling and the filling along paths to the lowest common ancestor of merge tree maps [9, 21]; extrema stay at their anchors, and each gap is then clipped or raised to its barrier b_k, which relaxed orders require. All frames share the direction, origin, extent and pixel scale, with no per-frame recentering. Because raster rounding and strict-extremum clearance can change the displayed merge levels, every rendered frame is checked against the original merge matrix, and records keep the requested budget, τ, δ^* and the measured errors. Capacity, input and solver failures are reported rather than repaired by rescaling.

6 Experiments

We evaluate whether anchoring improves absolute reference positioning, what topology relaxation changes, how accurate the position bounds are, and how the resulting maps support qualitative inspection. Four planar datasets provide the primary numerical evidence and two volumetric datasets add a local comparison with TMTM; the bounds are also tested on controlled trees and on the frames of an earlier benchmark. We distinguish published scores, local measurements and qualitative archives throughout.

Table 2: Balanced-mode parameter settings and runtime (seconds).

Parameter	Gaussian	Ring	Wildfires	Storms
ε	1%	1%	5%	10%
L	4096	196	4096	4096
s_w	0.00581	0.00328	0.00053	0.01170
t_{opt}	0.156	0.150	3.324	2.600
t_{total}	1.340	0.486	4.424	63.447

Weights $(4, 1, 0.5, 0.1)$ and $\rho = 0.5$ are fixed. s_w is width per support measure; L is raster length. Times are one run on Apple M4: optimization and full pipeline, respectively.

6.1 Datasets and Evaluation Protocol

Our four planar inputs are Gaussian, Ring, Wildfires and Storms (64, 40, 32 and 744 frames). Raster lengths are 4096, except Ring's original 196 positions. The ST-MTM persistence numbers are tested as per-frame range fractions $p = (0.001, 0.001, 0.15, 0.105)$, or $0.1\%, 0.1\%, 15\%, 10.5\%$; all 880 thresholds were checked [9]. Gaussian is a three-peak reconstruction, Ring uses the archived generator, Wildfires aggregates FRP into 32 grids, and Storms uses ERA5 anomalies with the archived moving-mean and smoothing convention. Author-processed arrays are unavailable, so these are disclosed local reconstructions, not identical-input reproductions.

The volumetric inputs are Cylinder (102 frames, $135 \times 64 \times 48$, Q-criterion) [7, 35] and Tangaroa (201 frames, $300 \times 180 \times 120$, velocity magnitude) [28]. Cylinder computes Q from the original $192 \times 64 \times 48$ velocity grids before cropping; 2% persistence uses each frame's scalar range, followed by one fixed sequence scaling of $-Q$. Tangaroa uses speed clipped to $[0, 2]$, replaces near-zero values below 10^{-6} by the archived inflow value 1, and retains ab-

Table 3: Four-dataset comparison. Bold/underline mark the best/second distinct local measurements per dataset; ties share emphasis. Curves use fixed default orientations.

Dataset	Hilbert			Morton			TMTM			ST-MTM			AnchorMTM (ours)			
	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	SNS†	TW†	NPE↓	SNS†	TW†	NPE↓	Mode	SNS↓	TW↑	NPE↓
Gaussian	0.1117	0.6719	0.7557	<u>0.0393</u>	<u>0.8333</u>	0.7359	0.2500	0.7700	<u>0.1118</u>	0.0600	1.0000	0.1154	Exact	**0.0353**	**0.9323**	**0.0089**
													Balanced	**0.0353**	**0.9323**	**0.0089**
Ring	**0.1252**	**0.9048**	0.5941	0.1489	0.7619	0.6033	0.2800	0.6800	0.1756	0.2100	0.8800	0.2463	Exact	0.1485	<u>0.8571</u>	<u>0.0471</u>
													Balanced	<u>0.1334</u>	<u>0.8571</u>	**0.0395**
Wildfires	0.1883	0.8303	0.3641	0.1304	0.8758	0.1560	0.4400	0.8800	0.2280	0.2400	0.9100	0.3180	Exact	**0.0426**	**0.9136**	<u>0.0131</u>
													Balanced	<u>0.0447</u>	<u>0.8989</u>	**0.0116**
Storms	0.1290	0.7403	0.3705	0.1180	0.7251	0.3468	0.3400	0.7100	0.5753	0.2100	0.9400	0.4579	Exact	<u>0.0790</u>	<u>0.8463</u>	<u>0.0399</u>
													Balanced	**0.0743**	**0.8766**	**0.0140**

†Grey: published TMTM/ST-MTM Temporal SNS/TW (ST-MTM Table 1, authors' inputs), shown for reference and excluded from emphasis; other entries are local measurements on the same inputs. Author inputs and coverage are unconfirmed. Local TW coverage (Gaussian/Ring/Wildfires/Storms) is 64/64, 1/40, 4/32, 11/744; ineligible frames are omitted. Published TW has two-decimal precision. Mode applies only to AnchorMTM.

Table 4: Volumetric datasets and Balanced-mode parameter settings.

Parameter	Cylinder	Tangaroa
Frames	102	201
Grid	$135 \times 64 \times 48$	$300 \times 180 \times 120$
Scalar	Q-criterion	Speed $\|\mathbf{v}\|$
Persistence	2% per-frame range	0.2 absolute
Reference	$+x$	$+x$
ε	1%	33.33%
L	32768	131072
s_w	2.70668×10^{-5}	0.40953
Weights	$(0.1, 16, 0, 0.1)$	$(0, 64, 0, 0)$
ρ	1	1
Gap/L_{world}	1/300	1/960
Extra/L_{world}	0.1	0.1

Weights are reference, geometry, temporal and eccentricity. s_w is width per support measure (Cylinder voxel counts; Tangaroa physical volume); L is raster length, distinct from the physical canvas extent L_{world}. Exact uses zero allowance with the same other parameters as its Balanced counterpart.

solute persistence 0.2. These preprocessing choices are shared with the local TMTM adapter. Cylinder's 102-frame subset differs from the 508 frames reported in the first paper; no author-score reproduction is claimed.

For the volumetric comparison, the reference is restricted to the positive Cartesian axes rather than the continuous direction in Section 3.3. The archived offline rule chooses the axis capturing the most squared displacement of matched support centroids, with a minimum-image correction for Cylinder's periodic z coordinate (period 6); both datasets select $+x$. Centroids select the axis only: placement targets remain leaf extrema projected onto that fixed axis and full-domain range. This rule uses the full input sequence, not output SNS/TW/NPE, and does not define an independent test. The stored layout configurations are exploratory selections; Table 4 reports their different weights and budgets. In particular, Tangaroa's zero reference weight makes its configuration geometry-dominated; low measured NPE is an outcome, not evidence that an absolute-reference penalty caused it.

Within each local dataset, methods share source trees, identities and geometry inputs, and NPE uses one common offline extremum reference and projected range. SNS measures squared scale-normalized stress between support-centroid and anchor distances [9, 30]; TW measures neighborhoods [33], with $k = (1, 3, 5, 3)$ and $0 < k < n_t/2$. Ineligible frames are omitted. Absolute anchoring uses NPE, the worst displacement $\max_i |x_i^t - q_i^t|$ of a frame divided by the physical extent L_{world} and averaged over frames. TMTM/ST-MTM SNS/TW are quoted from ST-MTM Table 1 (Temporal mode); their NPE is measured with local adapters and shared references, each mirrored once if its first multi-leaf frame correlates negatively with the reference. Table 3 distinguishes these sources. Local TW coverage is 64/64, 1/40, 4/32 and 11/744; author coverage is unspecified.

Hilbert and Morton provide sample-domain baselines [14]. Their grid orientation is fixed (x columns, y rows), with no rotation, reflection or reversal search. Curves use power-of-two padding, discard out-of-domain samples, and map each extremum's full-grid rank to the common canvas at the original raster length. They impose neither merge-tree topology nor finite-width constraints.

Mode selection. For the four planar datasets, Exact uses $\varepsilon = 0$. Balanced selects a positive budget from 1–10% (1% steps) on the first chronological quarter. Among legal reconstructions with attained $\mathrm{NMLE}_{\max} \le \varepsilon$, it minimizes $J = 0.5\,\mathrm{NPE}_{norm} + 0.5\,\mathrm{SNS}_{norm}$. Terms are min–max normalized over legal candidates; constant terms contribute zero. Within 0.01 of the minimum, smaller budgets then attained errors break ties. Budgets are frozen before full runs, without a zero-budget fallback (Supplementary Fig. S4). Full results include validation frames; earlier outputs were observed, so this is not a new blind test.

6.2 Reference Positioning and Spatial Quality

ST-MTM reports higher TW than AnchorMTM on all four inputs. Its published SNS values are higher than our local ones, but they come from the authors' inputs and frame coverage, so we do not rank SNS or TW across these sources; Wildfires Exact's TW of 0.9136 is also too close to the published two-decimal 0.91 to establish an advantage.

Against fixed Hilbert and Morton curves, Balanced improves SNS and TW on Gaussian, Wildfires and Storms. Ring is an exception: Hilbert achieves SNS 0.1252 and TW 0.9048, versus Balanced's 0.1334 and 0.8571; its TW evidence covers only one eligible frame. Balanced has lower NPE than both curves and both merge-tree adapters on every input. Reductions relative to the ST-MTM adapter are 92.3%, 84.0%, 96.4% and 96.9%. This supports alignment to the specified reference, not general visualization quality: traversal coordinates need not represent a physical projection, and relative geometry does not determine an absolute origin.

6.3 Volumetric Comparison with TMTM

Table 5 follows the planar table's metric order and Exact–Balanced grouping, but compares only with locally adapted TMTM. All three metrics are recomputed over the complete sequences; SNS uses three-dimensional support-centroid distances, TW uses $k = 1$ with full frame coverage, and NPE uses extremum targets along the common x axis. Balanced has lower SNS and NPE and higher TW than TMTM on both datasets. Exact and Balanced share the same axis, trees, retained leaf identities and layout weights; only the merge allowance differs. Exact has zero measured fixed-leaf merge error,

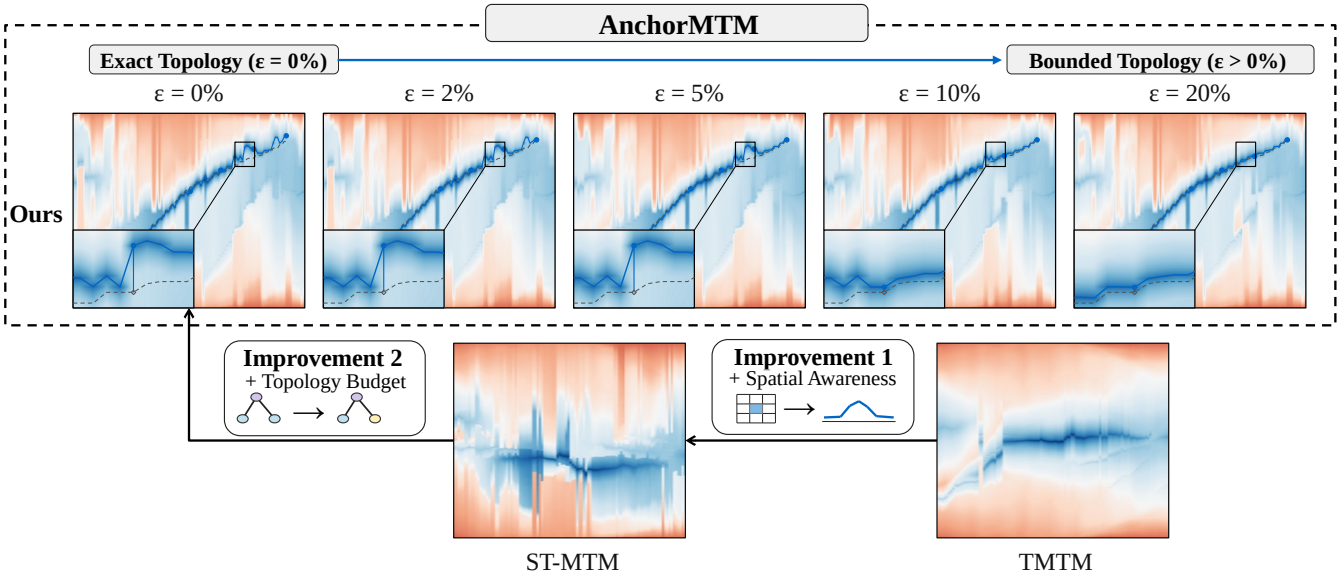


Figure 4: Storms parameter maps and method comparison. AnchorMTM spans exact topology ($\varepsilon = 0$) and bounded topology (2, 5, 10 and 20%), with the original trajectories and common local crops. TMTM and ST-MTM maps use the same scalar fields and augmented trees through local implementations. Spatial awareness and topology budget indicate the respective conceptual improvements. All maps show frames 355–460 with the original blue–orange scale, $[-35, 35]$ hPa. Module connections summarize design differences rather than exact algebraic equivalence; multiway-tree adapters are disclosed in the text.

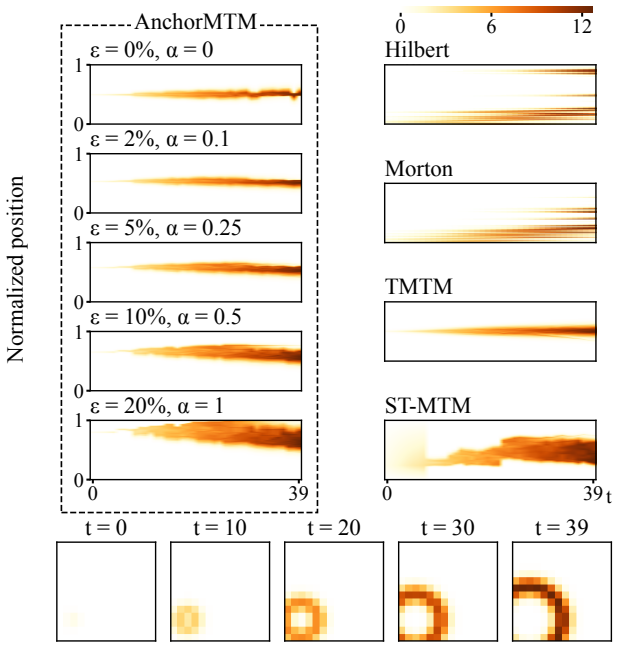


Figure 5: Ring comparison. Left: AnchorMTM with topology allowance ε and guidance strength $\alpha = \varepsilon/20\%$ raised together, from neutral central targets ($\alpha = 0$) to full spatial guidance ($\alpha = 1$); the upward shift reflects guidance, not the allowance alone. Right: supplied Hilbert/Morton reference panels and same-input local TMTM/ST-MTM results. Below: source snapshots at frames 0, 10, 20, 30 and 39. All computed maps use 40 frames and 196 positions; the scalar key applies to computed maps and snapshots. Vertical coordinates denote normalized layout position or curve rank.

Table 5: Volumetric comparison with a fixed x reference. Bold/underline mark best/second distinct values per dataset; ties share emphasis.

Dataset	TMTM			AnchorMTM (ours)			
	SNS↓	TW↑	NPE↓	Mode	SNS↓	TW↑	NPE↓
Cylinder	0.2956	<u>0.8017</u>	0.4769	Exact	<u>0.1226</u>	0.8010	<u>0.1773</u>
				Balanced	**0.0329**	**0.9117**	**0.0458**
Tangaroa	0.5255	0.6701	0.3866	Exact	<u>0.1204</u>	<u>0.8213</u>	<u>0.1285</u>
				Balanced	**0.1124**	**0.8408**	**0.0653**

All values are local measurements, including adapted TMTM. TW uses $k = 1$, with coverage 102/102 and 201/201. Mode applies only to AnchorMTM. Configurations are frozen exploratory results, not a blind test; settings appear in Table 4.

and both stored Balanced runs pass their declared topology budgets. The large Tangaroa allowance (33.33%) and its geometry-dominated objective limit attribution: these are selected configurations, not evidence of universal superiority or a controlled comparison with planar default weights.

6.4 Topology Validity and the Effect of Relaxation

Exact has zero fixed-leaf merge-matrix error. Balanced attains maximum NMLE of 0, 0.982%, 4.421% and 9.985%, within its allowances; all eight runs pass continuous and raster checks without collapsed anchors. Checks include diagonal extrema, not every augmented internal label. Ring requires checked joint integer rounding in 26 Exact and 21 Balanced frames, retaining its 196 positions and continuous order.

Balanced reduces NPE relative to Exact by 16.2% on Ring, 11.6% on Wildfires and 64.9% on Storms. Gaussian is unchanged despite its positive 1% budget. SNS improves on Ring and Storms but rises from 0.0426 to 0.0447 on Wildfires. Budget relaxation therefore does not improve every metric.

These Exact–Balanced comparisons measure the effect of the selected allowances, rather than a component ablation. A positive al-

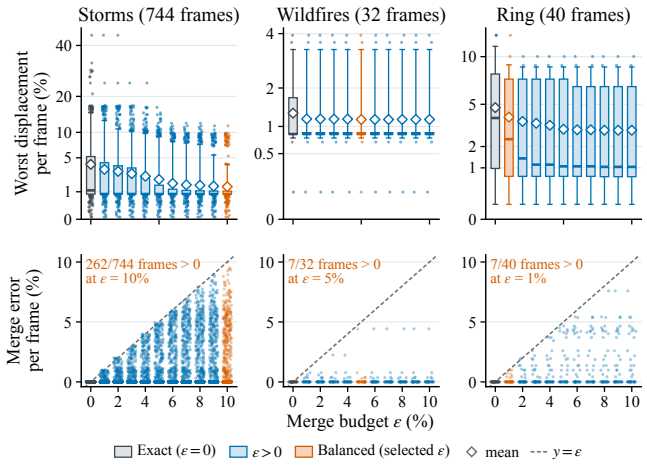


Figure 6: Per-frame distribution of AnchorMTM over the merge budget ε on the complete sequences, with the references, widths and weights of Table 3. Top: worst displacement per frame (square-root scale; boxes: quartiles; whiskers: 5th–95th percentiles; dots: other frames; diamonds: means, which give the table's NPE). Bottom: attained merge error per frame; no frame exceeds its budget. Gaussian is identical at every budget and omitted. Larger budgets, SNS and the frames involved appear in Supplementary Figs. S1 and S2.

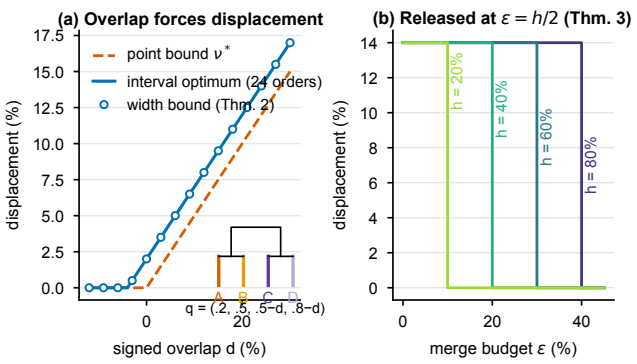


Figure 7: Controlled four-leaf trees (inset: sibling pairs and references). (a) Increasing sibling-range overlap d forces displacement; the width bound of Theorem 2 (circles) equals the optimum over all 24 leaf orders. (b) At $d = 24\%$, the 14% displacement disappears exactly when the merge budget reaches half the parent gap h, for all four gaps, as Theorem 3 predicts. Percentages refer to a unit canvas or a unit scalar range.

lowance need not change the selected order or improve reference positioning. The budget constrains merge-level error; it is not a direct bound on spatial displacement.

Figure 6 sweeps ε from 0 to 10% and shows that relaxation shortens the upper tail rather than shifting every frame. On Storms the median per-frame displacement moves only from 1.10% to 0.84% at $\varepsilon = 10\%$, while the 90th percentile falls from 10.7% to 2.5% and the worst frame from 44.8% to 11.5%; 482 of the 744 frames keep zero merge error. The frames that use the allowance form a few runs of consecutive frames (Supplementary Fig. S2). On Wildfires the same 7 of 32 frames use the allowance at every budget. On Ring the median falls from 3.9% to 1.0%, but the last frame stays at 10.0% until the budget reaches 63%. In the volumes under one default configuration, by contrast, almost every frame uses the allowance and the whole distribution moves (Supplementary Fig. S3).

6.5 Accuracy of the Position Bounds

Controlled trees. To separate the causes of displacement, we fix two sibling pairs $\{A,B\}$ and $\{C,D\}$ that merge at 0.4 below a root at 1, with leaf values 0 and references $(0.2, 0.5, 0.5-d, 0.8-d)$ on a unit canvas, and vary the signed overlap d from -0.12 to 0.30. Across 43 overlaps and four widths, we compare the bounds with all 24 leaf orders, each placed by an independent minimax LP. Topology alone forces exactly $\nu^* = \max(0,d)/2$. With widths 0.06, gap 0.01 and $\rho = 0.5$, the width bound, our search and the exhaustive optimum coincide at $\tau = \max(0,(d+0.04)/2)$ (Fig. 7a): 12% for points and 14% for intervals at $d = 0.24$. A merge allowance removes the conflict exactly where Theorem 3 admits interleaving, at $2\varepsilon = h$ for all four parent gaps h tested (Fig. 7b).

Real trees. We also evaluate the bounds on the frames of an earlier benchmark with one default configuration (Supplementary Material): five planar sequences with 614 frames, namely 12-hourly ERA5 sea-level pressure [19], daily wildfire radiative power [14], outgoing longwave radiation [22], $PM_{2.5}$ concentration [10] and Ring, and the Tangaroa and Cylinder volumes. In Exact Mode, the interval cost τ of the selected order equals $\mathscr{B}_w$ on 606 of the 614 planar frames, so on these frames no exact-topology order has a smaller worst displacement; at $\varepsilon = 10\%$ the bounded-topology certificate is met on 596. Volumetric trees are harder: the bound is met on 81 of 102 Cylinder and 27 of 201 Tangaroa frames, and the remaining gap may lie in the bound or in the search. Because Theorem 1 needs no layout, it also measures existing maps: midpoint ordering attains ν^* on the same tree, so order error above ν^* is chosen by the order rule rather than forced by the hierarchy. On these seven sequences, 53–95% of the order error of TMTM and 31–95% of that of ST-MTM lies above ν^*, each evaluated in the mirror that agrees better with the reference. Supplementary Fig. S5 places all maps against the bounds. Relaxation buys position where conflicting subtrees merge just below their parent: 1% halves the forced order error on Wildfires (5.80% to 2.77%), whereas on Tangaroa the bound stays above 7% up to $\varepsilon = 5\%$. Over 400 reference directions, the mean order bound spans 4.70–10.14% on Tangaroa and 5.49–28.15% on Cylinder, computed before any map is drawn (Supplementary Fig. S6).

6.6 Ring Case Study: Temporal Overview and Source Snapshots

Figure 5 shows a joint guidance–topology sweep on Ring: $\varepsilon = 0, 2, 5, 10, 20\%$ and guidance strength $\alpha = \varepsilon/20\%$. Layout targets interpolate between the canvas midpoint and the fixed $-128.7°$ physical projection: $q_{layout} = (1-\alpha)q_{mid} + \alpha q_{physical}$. Thus stronger guidance moves the reconstructed ring upward through actual optimization, with no image translation. This qualitative sweep changes both guidance and topology allowance, and each panel lists both values; it is not an epsilon-only ablation, its $\alpha = 0$ panel is not the Exact Mode of Table 3, and it does not replace the frozen main-table settings. TMTM/ST-MTM maps use the same fields and trees; Hilbert/Morton remain supplied references. Five source snapshots show expansion and strengthening.

6.7 Storms Case Study: Reading Budgeted Maps

The upper row of Fig. 4 retains the original five-budget qualitative Storms archive of Fig. 1, separate from main-table preprocessing. The 0, 2 and 5% layouts overlap locally, while 10 and 20% move the selected feature toward its fixed reference. The lower row shows TMTM and ST-MTM maps recomputed on these same 744 scalar fields and augmented trees, showing the same event window without per-panel recoloring or display translation. TMTM uses overlap-driven temporal subtree alignment in place of anchored placement; ST-MTM uses geometry-based ordering and distance-aware placement with temporal position constraints.

The local TMTM implementation handles genuine multiway saddles with equal-level separator copies; the ST-MTM adapter refines them into a stable-ID equal-level binary hierarchy and checks the original leaf LCA matrices. Both baselines use start frame 40 and 8192-position display rasters. They are local implementations, not author binaries. The module arrows describe replacements and additions rather than claiming exact algorithmic reductions or changing the main-table protocol.

6.8 Parameters and Computational Cost

Table 2 lists the four planar Balanced budgets (1, 1, 5 and 10%) and layout parameters; Table 4 separately lists the volumetric datasets and settings. The planar table reports one uncached Balanced run on Apple M4, with one BLAS/TTK thread, sequential extraction and the default HiGHS scheduler. Optimization includes ordering and placement; total time additionally includes preprocessing, trees, matching/reference estimation, rasterization, I/O and checks. Imports, reporting and metric export are excluded; OS caches may be warm. Timings do not establish speedup significance. Baseline NPE uses the disclosed CompactTMTM and ST-MTM solver adaptations, not author binaries.

7 Discussion and Limitations

The bounds separate forced from chosen displacement. Order error above v^* is always removable without changing the tree, since midpoint ordering attains v^*; the width-aware and bounded-topology bounds need not be attained, so an excess above them may lie in the bound rather than in the layout.

A fixed physical reference and a geometry-oriented embedding answer different questions. AnchorMTM supports reading positions along a chosen spatial coordinate; ST-MTM's higher reported TW highlights the value of preserving local neighborhoods. Neither property guarantees the other. The four-input results therefore support a task-specific anchoring claim, rather than universal superiority in spatial or temporal fidelity.

The empirical comparison has three limitations. First, author-processed fields and their small-sample TW protocol are not fully available, so published SNS/TW cannot provide an identical-input significance test. Second, the reference and tracking are computed offline from the supplied sequence; support overlap is a correspondence proxy, not a physical-identity annotation. Third, order search is heuristic and a continuous certificate does not remove integer-rendering constraints. We check actual fixed-label merge matrices after reconstruction, but do not claim a certificate for every augmented internal label.

The experiments also do not establish perceptual readability, physical-motion judgment accuracy, repeated-run speedups or generalization to unseen datasets. These require separate studies. Keeping source provenance, coverage, failed candidates and attained topology errors alongside the numerical scores is necessary when extending this evaluation.

8 Conclusion

Keeping the merge tree on the axis of a temporal merge tree map has a positional price; three layout-independent bounds state the displacement that exact topology, finite widths and a bounded merge error force on any layout. The selected exact-topology order attains the width-aware bound on 606 of 614 planar frames, and 31–95% of the order error of TMTM and ST-MTM is not forced by the tree. On four local inputs, AnchorMTM places features closer to their references than same-input baselines within every merge budget, while ST-MTM's published neighborhood preservation remains higher.

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